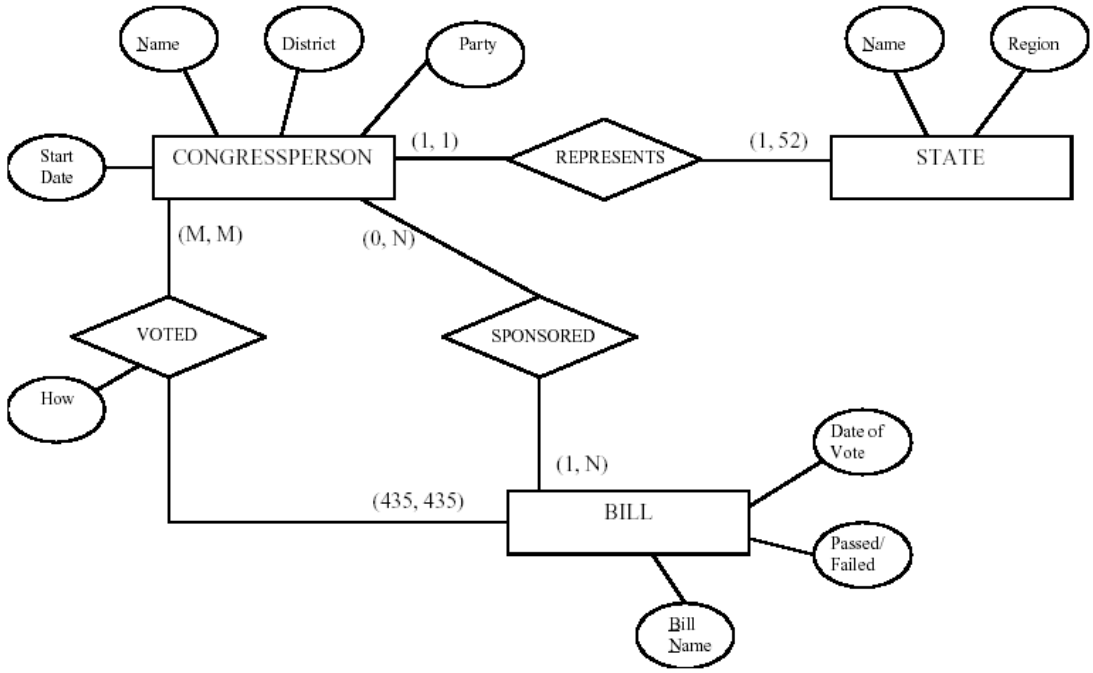
1. Design an ER schema for keeping track of information about votes taken in the U.S. House of Representatives during the current two-year congressional session. The database needs to keep track of each U.S. STATE's Name (e.g., 'Texas', 'New York', 'California') and include the Region of the state (whose domain is {'Northeast', 'Midwest', 'Southeast', 'Southwest', 'West'}). Each CONGRESS_PERSON in the House of Representatives is described by his or her Name, plus the District represented, the Start_date when the congressperson was first elected, and the political Party to which he or she belongs (whose domain is {'Republican', 'Democrat', 'Independent', 'Other'}). The database keeps track of each BILL (i.e., proposed law), including the Bill_name, the Date_of_vote on the bill, whether the bill Passed_or_failed (whose domain is {'Yes', 'No'}), and the Sponsor (the congressperson(s) who sponsored—that is, proposed—the bill). The database also keeps track of how each congressperson voted on each bill (domain of Vote attribute is {'Yes', 'No', 'Abstain', 'Absent'}). Draw an ER diagram for this application. State clearly any assumptions you make.



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2. Consider a MAIL_ORDER database in which employees take orders for parts from customers. The data requirements are summarized as follows:

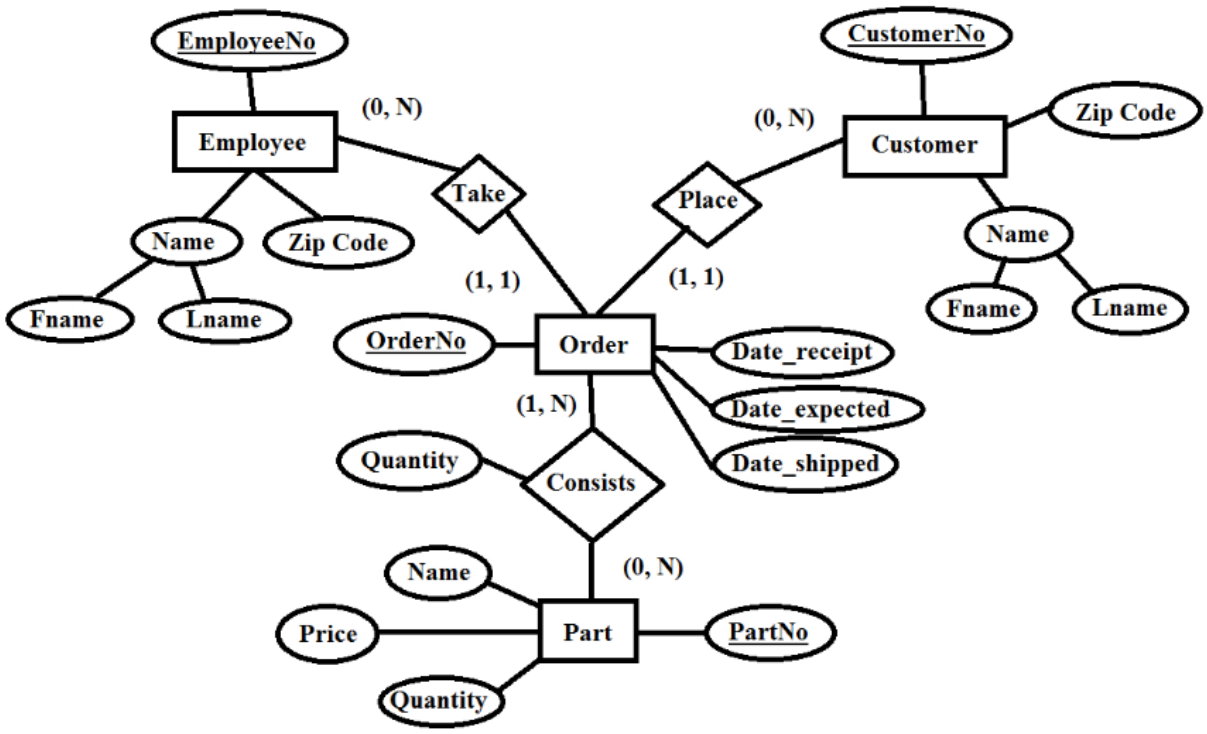
■ The mail order company has employees, each identified by a unique employee number, first and last name, and Zip Code.

■ Each customer of the company is identified by a unique customer number, first and last name, and Zip Code.

■ Each part sold by the company is identified by a unique part number, a part name, price, and quantity in stock.

■ Each order placed by a customer is taken by an employee and is given a unique order number. Each order contains specified quantities of one or more parts. Each order has a date of receipt as well as an expected ship date. The actual ship date is also recorded.

Design an entity–relationship diagram for the mail order database.



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3. Consider a CONFERENCE_REVIEW database in which researchers submit their research papers for consideration. Reviews by reviewers are recorded for use in the paper selection process. The database system caters primarily to reviewers who record answers to evaluation questions for each paper they review and make recommendations regarding whether to accept or reject the paper. The data requirements are summarized as follows:

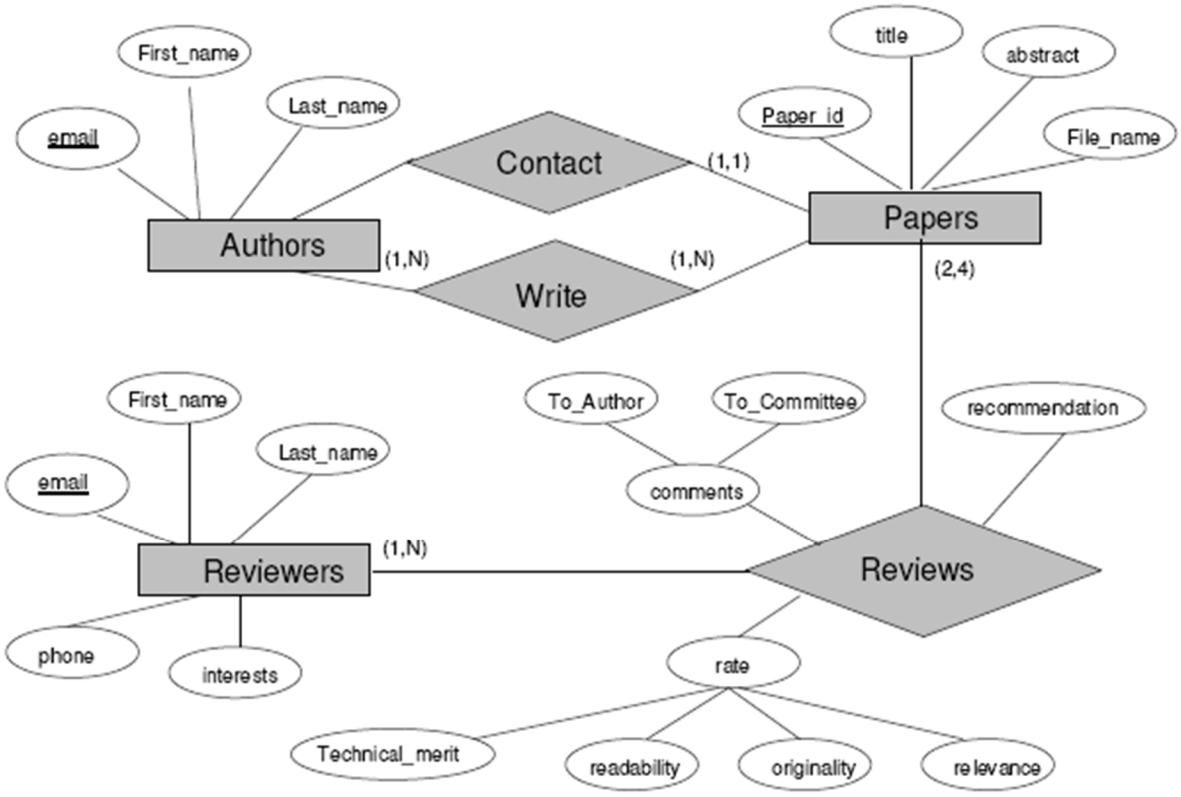
Authors of papers are uniquely identified by e-mail id. First and last names are also recorded.

Each paper is assigned a unique identifier by the system and is described by a title, abstract, and the name of the electronic file containing the paper.

A paper may have multiple authors, but one of the authors is designated as the contact author.

Reviewers of papers are uniquely identified by e-mail address. Each reviewer's first name, last name, phone number, affiliation, and topics of interest are also recorded.

Each paper is assigned between two and four reviewers. A reviewer rates each paper assigned to him or her on a scale of 1 to 10 in four categories: technical merit, readability, originality, and relevance to the conference. Finally, each reviewer provides an overall recommendation regarding each paper.

Each review contains two types of written comments: one to be seen by the review committee only and the other as feedback to the author(s).

Design an entity–relationship diagram for the CONFERENCE_REVIEW database.